

✧ **Properties on sets :**

(1) $A \cup A = A$ i.e., the union of sets is idempotent.

Proof : Let x be any arbitrary element of the set A then

$$x \in A \Rightarrow x \in A \text{ or } x \in A$$

$$\Rightarrow x \in A \cup A$$

$$\therefore A \subseteq A \cup A \quad \dots(i)$$

Again let y be any arbitrary element of set $A \cup A$, then

$$y \in A \cup A \Rightarrow y \in A \text{ or } y \in A$$

[by definition of union of sets]

$$\therefore A \subseteq A \cup A \quad \dots(ii)$$

From (i) and (ii), we get

$$A \cup A = A.$$

(2) $A \cup \phi = A$, where A is any set.

Proof : Let x be an arbitrary element of the set $A \cup \phi$. Then,

$$x \in A \cup \phi \Rightarrow x \in A \text{ or } x \in \phi$$

[by def. of union of sets]

$$\Rightarrow x \in A \quad [\because \phi \text{ is the null set}]$$

$$\therefore A \cup \phi \subseteq A \quad \dots(i)$$

Again let y be an arbitrary element of the set A , then

$$\begin{aligned}y \in A &\Rightarrow y \in A \text{ or } y \in \phi \\&\Rightarrow y \in A \cup \phi \\ \therefore A &\subseteq A \cup \phi \quad \dots(ii)\end{aligned}$$

From (i) and (ii), we get

$$A \cup \phi = A.$$

(3) If A is any subset of the universal set U , then $A \cup U = U$.

Proof : Let x be any arbitrary element of the $A \cup U$, then

$$\begin{aligned}x \in A \cup U &\Rightarrow x \in A \text{ or } x \in U \\&\Rightarrow x \in U \quad [\because A \text{ is the subset of } U] \\ \therefore A \cup U &\subseteq U \quad \dots(i)\end{aligned}$$

Again let y be any arbitrary element of the universal set U , then,

$$\begin{aligned}x \in U &\Rightarrow x \in U \text{ or } x \in A \\&\Rightarrow x \in A \cup U \\ \therefore U &\subseteq A \cup U \quad \dots(ii)\end{aligned}$$

From (i) and (ii), we have,

$$A \cup U = U.$$

(4) $A \cup A' = U$, where A is any subset of the universal set U and A' is its complement.

Proof : Every set is a subset of universal set.

$$\therefore A \cup A' \subseteq U \quad \dots(i)$$

Again let x be any arbitrary element of the universal set U . Since $A \subseteq U$. Therefore,

$$\begin{aligned}x \in U &\Rightarrow x \in A \text{ or } x \notin A \\&\Rightarrow x \in A \text{ or } x \in A' \\&\Rightarrow x \in A \cup A' \quad \dots(ii)\end{aligned}$$

Therefore, $U \subseteq A \subset A'$

From (i) and (ii), we get

$$A \cup A' = U.$$

(5) $A \cup B = B \cup A$, i.e., union of sets is commutative.

Proof : Let x be any arbitrary element of the set $A \cup B$, then

$$x \in A \cup B \Rightarrow x \in A \text{ or } x \in B$$

$$\Rightarrow x \in B \text{ or } x \in A$$

$$\Rightarrow x \in B \cup A$$

$$\therefore A \cup B \subseteq B \cup A \quad \dots(i)$$

Again let y be any arbitrary element of set $B \cup A$, then

$$y \in B \cup A \Rightarrow y \in B \text{ or } y \in A$$

$$\Rightarrow y \in A \text{ or } y \in B$$

$$\Rightarrow y \in A \cup B.$$

$$\therefore B \cup A \subseteq A \cup B \quad \dots(ii)$$

From (i) and (ii), we get,

$$A \cup B = B \cup A.$$

(6) $A \cup (B \cup C) = (A \cup B) \cup C$ where A , B and C are any three sets i.e., the union of sets is associative.

Proof : Let x be any arbitrary element of the set

$$A \cup (B \cup C) \text{ then } x \in A \cup (B \cup C)$$

$$\Rightarrow x \in A \text{ or } x \in (B \cup C)$$

$$\Rightarrow x \in A \text{ or } (x \in B \text{ or } x \in C)$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ or } x \in C$$

$$\Rightarrow x \in (A \cup B) \text{ or } x \in C$$

$$\Rightarrow x \in (A \cup B) \cup C$$

$$\therefore A \cup (B \cup C) \subseteq (A \cup B) \cup C \quad \dots(i)$$

Further let y be any arbitrary element of the set

$(A \cup B) \cup C$ then,

$$\begin{aligned}
 y \in (A \cup B) \cup C &\Rightarrow y \in (A \cup B) \text{ or } y \in C \\
 &\Rightarrow (y \in A \text{ or } y \in B) \text{ or } y \in C \\
 &\Rightarrow y \in A \text{ or } (y \in B \text{ or } y \in C) \\
 &\Rightarrow y \in A \text{ or } y \in (B \cup C) \\
 &\Rightarrow y \in A \cup (B \cup C)
 \end{aligned}$$

$$\therefore (A \cup B) \cup C \subseteq A \cup (B \cup C) \quad \dots(ii)$$

From (i) and (ii), we get,

or we conclude that,

$$A \cup (B \cup C) = (A \cup B) \cup C.$$

(7) $A \cap A = A$ i.e, the intersection of sets is idempotent.

Proof : Let x be any arbitrary element of the set $A \cap A$, then,

$$x \in A \cap A \Rightarrow x \in A \text{ and } x \in A$$

[by def. of intersection of sets]

$$\Rightarrow x \in A$$

$$\therefore A \cap A \subseteq A \quad \dots(i)$$

Again let y be any arbitrary element of the set A .

$$\text{Then, } y \in A \Rightarrow y \in A \text{ and } y \in A$$

$$\Rightarrow y \in A \cap A$$

$$\therefore A \subseteq A \cap A \quad \dots(ii)$$

From (i) and (ii), we have,

$$A \cap A = A.$$

(8) $A \cap \phi = \phi$.

Proof : Suppose $A \cap \phi \neq \phi$.

Let x be any arbitrary element of the set $A \cap \phi$, then,

$$x \in A \cap \phi \Rightarrow x \in A \text{ and } x \in \phi$$

[by def. of intersection of sets]

$$\Rightarrow x \in \phi$$

which is wrong.

$\therefore \phi$ has no element.

$\therefore$ Our supposition that $A \cap \phi \neq \phi$ is wrong.

$\therefore A \cap \phi$ has no element.

$\therefore A \cap \phi = \phi$.

(9) If A is any subset of the universal set U , then, $A \cap U = A$.

Proof : Let x be any arbitrary element of the set $A \cap U$, then

$$\begin{aligned}x \in A \cap U &\Rightarrow x \in A \text{ and } x \in U \\&\Rightarrow x \in A\end{aligned}$$

$$\therefore A \cap U \subseteq A \quad \dots(i)$$

Again let y be any arbitrary element of the set A , then

$$\begin{aligned}y \in A &\Rightarrow y \in A \text{ and } y \in U \\&\Rightarrow y \in A \cap U\end{aligned}$$

$$\therefore A \subseteq A \cap U \quad \dots(ii)$$

From (i) and (ii), we have,

$$A \cap U = A.$$

(10) $A \cap A' = \phi$.

Proof : Suppose $A \cap A' \neq \phi$.

Let x be any arbitrary element of the set $A \cap A'$ then,

$$\begin{aligned}x \in A \cap A' &\Rightarrow x \in A \text{ and } x \in A' \\&\Rightarrow x \in A \text{ and } x \notin A\end{aligned}$$

which is false.

$\therefore$ Our supposition is wrong.

$\therefore A \cap A' = \phi$.

(11) If A and B are any sets, then $A \cap B = B \cap A$ i.e., Intersection of sets is commutative.

Proof : Let x be any arbitrary element of the set $A \cap B$, then

$$x \in A \cap B \Rightarrow x \in A \text{ and } x \in B$$

[By def. of intersection of sets]

$$\Rightarrow x \in B \text{ and } x \in A$$

$$\Rightarrow x \in B \cap A$$

$$\therefore A \cap B \subseteq B \cap A \quad \dots(i)$$

Again let y be any arbitrary element of the set $B \cap A$, then

$$y \in B \cap A \Rightarrow y \in B \text{ and } y \in A$$

$$\Rightarrow y \in A \text{ and } y \in B$$

$$\Rightarrow y \in (A \cap B)$$

$$\therefore B \cap A \subseteq A \cap B \quad \dots(ii)$$

From (i) and (ii), we have

$$A \cap B = B \cap A.$$

(12) If A , B and C are any three sets, then,

$$A \cap (B \cap C) = (A \cap B) \cap C.$$

Proof : Let x be any arbitrary element of the set

$$A \cap (B \cap C) \text{ then } x \in A \cap (B \cap C)$$

$$\Rightarrow x \in A \text{ and } x \in (B \cap C)$$

$$\Rightarrow x \in A \text{ and } (x \in B \text{ and } x \in C)$$

$$\Rightarrow (x \in A \text{ and } x \in B) \text{ and } x \in C$$

$$\Rightarrow x \in (A \cap B) \text{ and } x \in C$$

$$\Rightarrow x \in (A \cap B) \cap C$$

$$\therefore A \cap (B \cap C) \subseteq (A \cap B) \cap C \quad \dots(i)$$

Again let y be any arbitrary element of the set $(A \cap B) \cap C$,

$$\text{then } y \in (A \cap B) \cap C \Rightarrow y \in (A \cap B) \text{ and } y \in C$$

$$\Rightarrow (y \in A \text{ and } y \in B) \text{ and } y \in C$$

$$\Rightarrow y \in A \text{ and } (y \in B \text{ and } y \in C)$$

$$\Rightarrow y \in A \text{ and } y \in (B \cap C)$$

$$\Rightarrow y \in A \cap (B \cap C)$$

$$\therefore (A \cap B) \cap C \subseteq A \cap (B \cap C) \quad \dots(ii)$$

From (i) and (ii), we have,

$$A \cap (B \cap C) = (A \cap B) \cap C.$$

✧ **Distributive laws :**

(1) If A, B, C are any three sets, then

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

i.e., The union of sets distributes over the intersection of sets.

Proof : Let x be any arbitrary element of the set,

$$A \cup (B \cap C) \text{ then } x \in A \cup (B \cap C)$$

$$\Rightarrow x \in A \text{ or } x \in (B \cap C)$$

$$\Rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C)$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C)$$

$$\Rightarrow x \in A \cup B \text{ and } x \in A \cup C$$

$$\Rightarrow x \in (A \cup B) \cap (A \cup C)$$

$$\therefore A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \quad \dots(i)$$

Again let y be any arbitrary element of the set,

$$(A \cup B) \cap (A \cup C)$$

$$\text{Then } y \in (A \cup B) \cap (A \cup C)$$

$$\Rightarrow y \in (A \cup B) \text{ and } y \in (A \cup C)$$

$$\Rightarrow (y \in A \text{ or } y \in B) \text{ and } (y \in A \text{ or } y \in C)$$

$$\Rightarrow y \in A \text{ or } (y \in B \text{ and } y \in C)$$

$$y \in A \text{ or } y \in (B \cap C) \Rightarrow y \in A \cup (B \cap C)$$

$$\therefore (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C) \quad \dots(ii)$$

From (i) and (ii), we have,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

(2) If A, B, C are any three sets, then

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

i.e., the intersection of sets distributes over the union of sets.

Proof : Let x be any arbitrary element of the set,

$$A \cap (B \cup C).$$

$$\begin{aligned}
 \text{Then, } x \in A \cap (B \cup C) &\Rightarrow x \in A \text{ and } x \in (B \cup C) \\
 &\Rightarrow x \in A \text{ and } (x \in B \text{ or } x \in C) \\
 &\Rightarrow (x \in A \text{ and } x \in B) \text{ or} \\
 &\quad (x \in A \text{ and } x \in C) \\
 &\Rightarrow x \in (A \cap B) \text{ or } x \in (A \cap C) \\
 &\Rightarrow x \in (A \cap B) \cup (A \cap C)
 \end{aligned}$$

$$\therefore A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C) \quad \dots(i)$$

Again let y be any arbitrary element of the set,

$$(A \cap B) \cup (A \cap C)$$

$$\text{Then, } y \in (A \cap B) \cup (A \cap C)$$

$$\begin{aligned}
 &\Rightarrow y \in (A \cap B) \text{ or } y \in (A \cap C) \\
 &\Rightarrow (y \in A \text{ and } y \in B) \text{ or } (y \in A \text{ and } y \in C) \\
 &\Rightarrow y \in A \text{ and } (y \in B \text{ or } y \in C) \\
 &\Rightarrow y \in A \text{ and } y \in (B \cup C) \\
 &\Rightarrow y \in A \cap (B \cup C)
 \end{aligned}$$

$$\therefore (A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C) \quad \dots(ii)$$

From (i) and (ii), we have,

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$